

3 Moments of inertia of a rigid body

What students need to learn:

Moments of inertia and radius of gyration of standard and composite bodies.

Use of integration including the proof of the standard results given in the formulae booklet will be required.

The parallel and perpendicular axes theorems.

4 Rotation of a rigid body about a fixed smooth axis

What students need to learn:

Motion of a rigid body about a fixed smooth horizontal or vertical axis.

Use of conservation of energy will be required. Calculation of the force on the axis will be required.

Angular momentum.

Kinetic energy.

Conservation of angular momentum. The effect of an impulse on a rigid body which is free to rotate about a fixed axis.

Simple pendulum and compound pendulum.

Unit S1

Statistics 1

GCE AS and GCE Mathematics, GCE AS and GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional) AS optional unit

S1.1 Unit description

Mathematical models in probability and statistics; representation and summary of data; probability; correlation and regression; discrete random variables; discrete distributions; the Normal distribution.

S1.2 Assessment information

Examination

The examination will consist of one 1½ hour paper. The paper will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know formulae which might be required by the specification and which are not included in the booklet, Mathematical Formulae including Statistical Formulae and Tables, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

$$\text{Mean} = \bar{x} = \frac{\sum x}{n} \text{ or } \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{(\text{Variance})}$$

$$\text{Interquartile range} = \text{IQR} = Q_3 - Q_1$$

$$P(A') = 1 - P(A)$$

For independent events A and B ,

$$P(B|A) = P(B), P(A|B) = P(A),$$

$$P(A \cap B) = P(A) P(B)$$

$$E(aX + b) = aE(X) + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Cumulative distribution function for a discrete random variable:

$$F(x_0) = P(X \leq x_0) = \sum_{x \leq x_0} p(x)$$

$$\text{Standardised Normal Random Variable } Z = \frac{X - \mu}{\sigma}$$

where $X \sim N(\mu, \sigma^2)$

1 Mathematical models in probability and statistics

What students need to learn:

The basic ideas of mathematical modelling as applied in probability and statistics.

2 Representation and summary of data

What students need to learn:

Histograms, stem and leaf diagrams, box plots.

Using histograms, stem and leaf diagrams and box plots to compare distributions.

Back-to-back stem and leaf diagrams may be required.

Drawing of histograms, stem and leaf diagrams or box plots will not be the direct focus of examination questions.

Measures of location — mean, median, mode.

Calculation of mean, mode and median, range and interquartile range will not be the direct focus of examination questions.

Students will be expected to draw simple inferences and give interpretations to measures of location and dispersion. Significance tests will not be expected.

Data may be discrete, continuous, grouped or ungrouped. Understanding and use of coding.

Measures of dispersion — variance, standard deviation, range and interpercentile ranges.

Simple interpolation may be required. Interpretation of measures of location and dispersion.

Skewness. Concepts of outliers.

Students may be asked to illustrate the location of outliers on a box plot. Any rule to identify outliers will be specified in the question.